

Mathematical preliminaries

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7 Exercises

All dimensions below are positive integers.

1. For $x > 0$, calculate

$$\frac{d}{dx} \left(4x^{5/2} + x^{1/2} - 6x^{-1/2} \right).$$

2. Use the chain rule to calculate

$$\frac{d}{dx} (2x + 5)^3.$$

Name the inner and outer functions and indicate where their derivatives are evaluated.

3. Calculate the same derivative by expanding $(2x + 5)^3$ first. Check that the answers agree.
 4. Find the global minimiser and the minimum value of

$$F(a) := 4a^2 + 2a + 1, \quad a \in \mathbb{R}.$$

Justify that the stationary point you find is indeed a global minimiser.

5. Calculate $\sum_{i=0}^5 i^2$.

6. Calculate $\prod_{i=1}^5 i$.

7. Simplify $\sum_{i=1}^N b$, where b does not depend on i .

8. Simplify $\prod_{i=1}^N b$, where b does not depend on i .

9. Compute

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$$

Compare the result with the product in Section 3.3, where the order was reversed.

10. Check invertibility and compute the inverse of

$$\mathbf{A} := \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

11. Use the inverse from Exercise 10 to solve

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Verify your solution by substitution.

12. Let $\mathbf{X} \in \mathbb{R}^{N \times d}$ be fixed and define the vector-valued function

$$\mathbf{g} : \mathbb{R}^d \longrightarrow \mathbb{R}^N, \quad \mathbf{g}(\mathbf{w}) := \mathbf{X}\mathbf{w}.$$

Write $\mathbf{w} = (w_1, \dots, w_d)^\top$. Its i th output component is the scalar function g_i . For $1 \leq i \leq N$ and $1 \leq k \leq d$, find

$$\partial_k g_i(\mathbf{w}) = \left(\frac{\partial g_i}{\partial w_k} \right) (\mathbf{w})$$

in terms of the entries of $\mathbf{X}$. Here $\mathbf{w}$ and $\mathbf{g}(\mathbf{w})$ are columns. Begin by writing $g_i(\mathbf{w})$ as a sum, and explain which quantities remain fixed during differentiation. State whether the derivative you obtain is a scalar, a vector or a matrix.

13. Let

$$F(a, b) := a^2 + ab + 2b^2.$$

Find its two partial derivatives and evaluate them at $(1, 2)$. With a column $\mathbf{w} = (w_1, w_2)^\top$, define $Q(\mathbf{w}) := F(w_1, w_2)$ and give its gradient at $\mathbf{w} = (1, 2)^\top$. State what is held fixed in each partial derivative.

14. Define

$$H(a, b) := (2a - b + 1)^2.$$

Use the scalar chain rule to calculate both partial derivatives. Identify the inner and outer functions, and evaluate the derivatives at $(1, 2)$. Now define

$$K(a, b) := \log(2a - b + 1)$$

on the domain $2a - b + 1 > 0$. Calculate its two partial derivatives using the same inner function.

15. Consider the function

$$F(s) := \log(1 + e^s), \quad s \in \mathbb{R}. \tag{36}$$

Explain why the function in (36) is defined for every real s . Use the chain rule to calculate $F'(s)$, identifying the inner and outer functions and where their derivatives are evaluated. Find $F'(0)$.

Solutions

1. On $x > 0$, the power rule gives

$$\begin{aligned}\frac{d}{dx} \left(4x^{5/2} + x^{1/2} - 6x^{-1/2} \right) &= 4 \left(\frac{5}{2} x^{3/2} \right) + \frac{1}{2} x^{-1/2} - 6 \left(-\frac{1}{2} x^{-3/2} \right) \\ &= 10x^{3/2} + \frac{1}{2} x^{-1/2} + 3x^{-3/2}.\end{aligned}$$

2. Define the inner and outer functions by

$$u(x) = 2x + 5, \quad v(s) = s^3, \quad F(x) = v(u(x)) = (2x + 5)^3.$$

Their derivatives are

$$u'(x) = 2, \quad v'(s) = 3s^2.$$

Applying the chain rule (22) gives

$$F'(x) = v'(u(x)) u'(x) = 3(2x + 5)^2 \cdot 2 = 6(2x + 5)^2, \quad x \in \mathbb{R}. \quad (37)$$

In (37), the derivative of the outer function is evaluated at $s = u(x) = 2x + 5$, while the derivative of the inner function is evaluated at x .

3. Expanding the cube first gives

$$(2x + 5)^3 = (2x)^3 + 3(2x)^2 \cdot 5 + 3(2x) \cdot 5^2 + 5^3 = 8x^3 + 60x^2 + 150x + 125. \quad (38)$$

Differentiating the polynomial in (38) term by term gives

$$\begin{aligned}\frac{d}{dx} (2x + 5)^3 &= 24x^2 + 120x + 150 \\ &= 6(4x^2 + 20x + 25) = 6(2x + 5)^2.\end{aligned}$$

This agrees with the chain-rule result (37).

4. For the objective in Exercise 4,

$$F'(a) = 8a + 2, \quad F'(a) = 0 \iff a = -\frac{1}{4}. \quad (39)$$

To check that the stationary point in (39) is a global minimiser, complete the square:

$$F(a) = 4a^2 + 2a + 1 = 4 \left(a + \frac{1}{4} \right)^2 + \frac{3}{4}. \quad (40)$$

The squared term in (40) is nonnegative for every $a \in \mathbb{R}$ and equals zero only at $a = -1/4$. Therefore the global minimiser is unique, and

$$\boxed{a^* = -\frac{1}{4}, \quad \min_{a \in \mathbb{R}} F(a) = F(a^*) = \frac{3}{4}.}$$

5. Including both limits gives $\sum_{i=0}^5 i^2 = 0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$.
 6. Multiplying the five factors gives $\prod_{i=1}^5 i = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$.
 7. There are N identical summands, so $\sum_{i=1}^N b = Nb$.
 8. There are N identical factors, so $\prod_{i=1}^N b = b^N$.
 9. Multiplying rows by columns gives

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}.$$

This is not $\begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}$, the result with the order reversed. The two products exist and have the same shape, but differ in value.

10. The determinant is $1 \cdot 4 - 2 \cdot 3 = -2 \neq 0$, so the matrix is invertible. Formula (11) gives

$$\mathbf{A}^{-1} = -\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}.$$

For example,

$$\mathbf{A}^{-1}\mathbf{A} = \begin{pmatrix} -2+3 & -4+4 \\ 3/2-3/2 & 3-2 \end{pmatrix} = \mathbf{I}_2.$$

11. Writing the system as $\mathbf{A}\mathbf{u} = \mathbf{c}$ with $\mathbf{u}, \mathbf{c}$ columns, the inverse gives

$$\mathbf{u} = \mathbf{A}^{-1}\mathbf{c} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -10+6 \\ 15/2-3 \end{pmatrix} = \begin{pmatrix} -4 \\ 9/2 \end{pmatrix}.$$

Substitution verifies the result:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -4 \\ 9/2 \end{pmatrix} = \begin{pmatrix} -4+9 \\ -12+18 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

12. Fix an output index i and a weight index k . Matrix multiplication gives

$$g_i(\mathbf{w}) = \sum_{j=1}^d X_{ij}w_j.$$

The upper limit is d , the number of columns of $\mathbf{X}$, because $\mathbf{w}$ has d coordinates. We vary w_k , keep all other coordinates w_j fixed, and keep every entry of $\mathbf{X}$ fixed. By linearity and the coordinate derivative rule,

$$\frac{\partial g_i}{\partial w_k}(\mathbf{w}) = \sum_{j=1}^d X_{ij} \frac{\partial w_j}{\partial w_k} = \sum_{j=1}^d X_{ij} \delta_{jk} = X_{ik}.$$

The derivative is a scalar and does not depend on the current value of $\mathbf{w}$.

13. Holding b fixed in the first calculation and a fixed in the second gives

$$\partial_1 F(a, b) = 2a + b, \quad \partial_2 F(a, b) = a + 4b.$$

At $(1, 2)$, these are 4 and 9. Since the argument of Q is a column,

$$\nabla Q \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix} \right) = \begin{pmatrix} 4 \\ 9 \end{pmatrix}.$$

14. For H , the inner function is $u(a, b) = 2a - b + 1$ and the outer function is $q(s) = s^2$. Their derivatives are

$$\partial_1 u(a, b) = 2, \quad \partial_2 u(a, b) = -1, \quad q'(s) = 2s.$$

Holding the other argument fixed and applying the scalar chain rule gives

$$\begin{aligned} \partial_1 H(a, b) &= q'(u(a, b)) \cdot 2 = 4(2a - b + 1), \\ \partial_2 H(a, b) &= q'(u(a, b)) \cdot (-1) = -2(2a - b + 1). \end{aligned}$$

Since $u(1, 2) = 1$, the two derivatives at $(1, 2)$ are 4 and -2 .

For K , the outer function is $p(s) = \log s$, so $p'(s) = 1/s$ on $s > 0$. On the stated domain,

$$\partial_1 K(a, b) = \frac{2}{2a - b + 1}, \quad \partial_2 K(a, b) = -\frac{1}{2a - b + 1}.$$

15. By (1), $e^s > 0$ for every real s , so $1 + e^s > 1 > 0$. The logarithm in (36) therefore has a positive argument for every $s \in \mathbb{R}$.

Define the inner and outer functions by

$$u(s) = 1 + e^s, \quad v(z) = \log z \quad (z > 0), \quad F = v \circ u.$$

The rules (20) give

$$u'(s) = e^s, \quad v'(z) = \frac{1}{z}.$$

Applying the chain rule (22), we obtain

$$F'(s) = v'(u(s)) u'(s) = \frac{1}{1 + e^s} e^s = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}. \quad (41)$$

In (41), the outer derivative is evaluated at $z = u(s) = 1 + e^s$; the inner derivative is evaluated at s . At $s = 0$, the same formula gives

$$F'(0) = \frac{e^0}{1 + e^0} = \frac{1}{2}.$$